

Lec_31

$\Rightarrow$ DAC $\Rightarrow$ min-max

Practice Questions of GATE

no of elements = 150

1) In an array to find min and max, find min number of comparison in best and worst case.

BC
w/o DPC

BC $\Rightarrow 1 \cdot (n-1)$
AC $\Rightarrow 1.5(n-1)$
WC $\Rightarrow 2(n-1)$

$$1) BC = (150 - 1) = 149$$

$$2) WC = 1.5 \times 150 - 2$$

with DPC
 $1.5n - 2$

$1.5n - 2$
 $\rightarrow$ each case

Car

2.) TC to get element from array which is neither max nor min? $\rightarrow O(1)$

array = [2, 10, 5, 6, 4, 27, 29, 1] $\rightarrow O(1)$
 $\checkmark \checkmark \checkmark \checkmark \checkmark \checkmark \checkmark$
max = 29, min = 1

take any 3
 or
 selecting 3
 elements sort
 them and
 return middle
 element

TC?

[2, 10, 5] $\rightarrow$ sorted
 $\downarrow$
max = 10, min = 2
Ans = 5

take 3 elements (any)

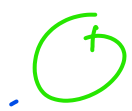
sorted then

$\Rightarrow$ sort $\rightarrow 3 \log 3$
 $\Rightarrow$ return $a(1) \rightarrow$

$n \log n \rightarrow$ merge sort
 $3 \log 3 \rightarrow$ constant

$3 \log 3 + 6$
 $O(1)$
 let array is

(2, 10, 5) $\Rightarrow$ [2, 5, 10]
 $a[0] = \min, a[2] = \max$



[15, 11, 1, 3, 6, 9]
max min

(1) (11) (15)

3.) TC to get element from array which is neither 2nd max nor 2nd min?

fmax = 10
smax = 5

fmin = 1
smin = 2

Time
5x0.5
+
O(1)

O(1) [10, 5, 1, 2, 3]

1) take 5 Elements
[(1, 2), 3, (5, 10)]

So get them
return middle

(2, 5, 10, 6, 9) 27 29 1 4 3 32
fmax = 32
smax = 29
fmin = 1
smin = 2

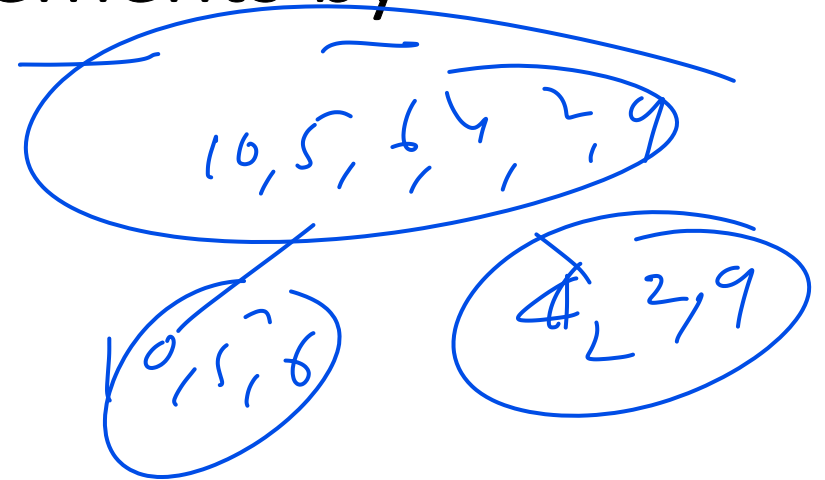
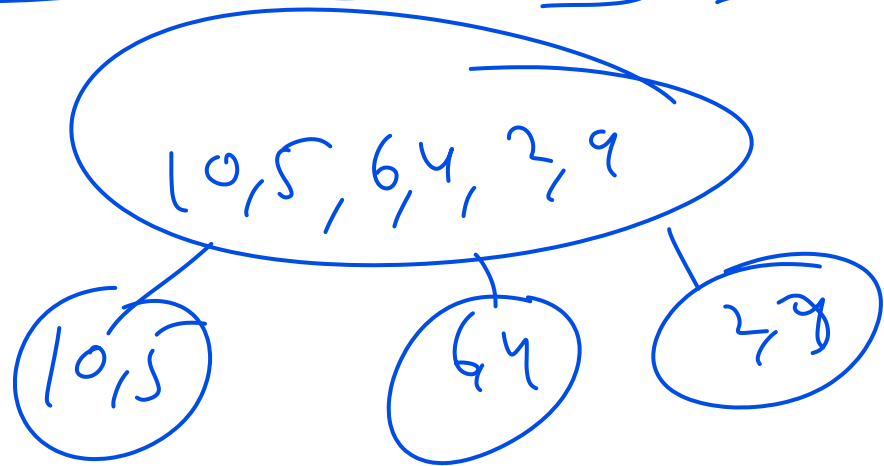
Note :) TC to get element from array which is neither kth max nor kth min?

$n \log n$
 $TC = k \log k$

take $2k+1$ element
sort them $\rightarrow (2k+1) \log(2k+1)$
 $\rightarrow k \log k + C$
 $= k \log k$

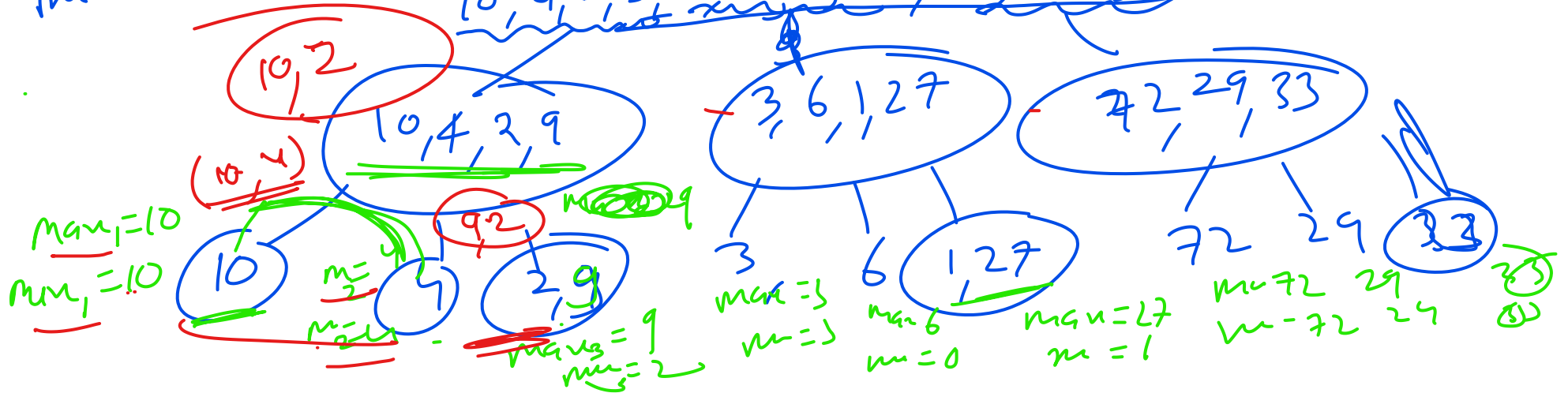
1st max $\rightarrow 3$
2nd max $\rightarrow 2$
3rd max $\rightarrow 5$

4.) Using DAC find min max elements by dividing array in 3 parts



max?
min?

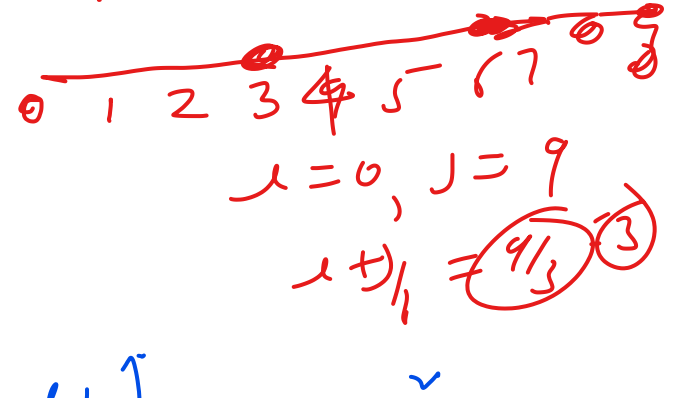
0 1 2 3 4 5 6 7 8 9 10
10, 4, 2, 9, 3, 6, 1, 2, 7, 7, 2, 2, 9, 33



$DAC(a, i, j)$
 $\{ \text{if}(i == j) \{ \text{max} = \text{min} = a(i) \}$
 $\{ \text{if}(i == j-1) \{ \text{if}(a(i) > a(j)) \{ \text{max} = a(i), \text{min} = a(j) \}$
 $\{ \text{else} \{ \text{max} = a(j), \text{min} = a(i) \}$
 $\{ \text{return}(\text{max}, \text{min}) \}$

else
 $\{ \text{mid} = (i+j)/3$
 $\{ \text{max}_1, \text{min}_1 = DAC(a, i, \text{mid})$
 $\{ \text{max}_2, \text{min}_2 = DAC(a, \text{mid}+1, j)$
 $\{ \text{max}_3, \text{min}_3 = DAC(a, 2\text{mid}+1, j)$

$\{ \text{if}(\text{max}_1 > \text{max}_2) \{ \text{max} = \text{max}_1 \}$
 $\{ \text{else} \{ \text{max} = \text{max}_2 \}$
 $\{ \text{if}(\text{max}_3 > \text{max}) \{ \text{max} = \text{max}_3 \}$
 $\{ \text{if}(\text{min}_1 < \text{min}_2) \{ \text{min} = \text{min}_1 \}$
 $\{ \text{else} \{ \text{min} = \text{min}_2 \}$



$$TC = 3T(n/3) + C$$

if $\min_3 \min_2 \& \min = \min_3$
 return $(\max, \min)$

no. of comparisons $T(n)$

$n=1$
 $n=2$

$$3T(n/3) + 4$$

$$T(n) = 3T(n/3) + 4$$

$\swarrow \searrow$
 $1/3$

$$T(n) = 2T(n/2) + 2$$

$\swarrow \searrow$
 2 prob

(2) power of cm

else

